

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.3 THE PERIODIC TABLE: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have now finished the Periodic Table unit, starting with the Mystery Element Card Challenge and ending with the Malindi salt case study. Your job is to write a complete, evidence-based explanation that answers the Driving Question: Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do? Use your Driving Question Board, your Model Versions 1 to the final poster, your trend tables, dot-and-cross diagrams and balanced equations. Write in complete sentences, use scientific words correctly (group, period, outer-shell electrons, cation, anion, valency, oxidation number), and give specific evidence from the lessons for every claim. Write your answers in the space under each prompt. Remember that explaining WHY matters more than listing facts.

Part 1: The Mystery Element Cards and the History of the Periodic Table

In Lesson 1 your class sorted 20 unlabelled element cards (H to Ca) and produced different groupings. (a) Describe two different ways groups sorted the cards and the puzzle the class could not yet explain, using the pairs Li/Na, F/Cl and He/Ne as evidence. (b) Using Lesson 2, explain how Döbereiner, Newlands, Mendeleev and Moseley each tried to organise the elements, and why ordering by atomic number worked better than ordering by mass. (c) Explain how the scientists' strategies were similar to your own card sorting.

(a) In Lesson 1 some groups sorted the cards by atomic number or mass, some by reactivity rating, some by state at room temperature, and some by the Kenyan context on the card (for example sodium in salt for ugali, chlorine in water treatment at Gigiri, calcium in Maasai cattle bones, carbon in charcoal for nyama choma). The puzzle was that elements far apart in atomic number, such as Li and Na, F and Cl, or He and Ne, had very similar properties, while elements sitting next to each other could be completely different. We could not explain this because we could not see what was inside the atoms. (b) Döbereiner (1817) grouped elements with similar properties in triads. Newlands (1864) arranged them by mass in a Law of Octaves. Mendeleev (1869) ordered them by atomic mass but left brave gaps for undiscovered elements and used them to predict properties. Moseley (1913) ordered them by atomic number, the number of protons, which fixed all the remaining anomalies and showed a deeper, invisible pattern inside the atom. Ordering by atomic number works better because it is the real identity of each element, whereas mass can put some elements in the wrong place. (c) Like the scientists, we tried different ordering rules (mass, reactivity, appearance), compared them, disagreed, and kept the rule that best put similar elements together. The periodic table developed over nearly 100 years as people proposed, tested and revised ideas, just as our class did. Our early sorting could only describe patterns; it could not yet explain them.

Part 2: Electron Arrangement and Position in the Table

Using Lesson 3, (a) write the

(a) Na has 11 electrons: $1s^2 2s^2 2p^6 3s^1$ (2,8,1). Cl has 17 electrons: $1s^2 2s^2 2p^6 3s^2 3p^5$ (2,8,7). Ca has 20 electrons: $1s^2 2s^2$

electron configurations in s/p notation (as you did in class) for Na, Cl, Ca, K and Ar. (b) Explain what the group number and the period number tell you about an atom. (c) Use this to explain why Na and K behave alike, and why Cl (Gigiri water treatment) and Ar (welding) behave so differently even though they are neighbours.	2p6 3s2 3p6 4s2 (2,8,8,2). K has 19 electrons: 1s2 2s2 2p6 3s2 3p6 4s1 (2,8,8,1). Ar has 18 electrons: 1s2 2s2 2p6 3s2 3p6 (2,8,8). (b) The group number tells you the number of outer-shell (valence) electrons, and the period number tells you how many electron shells are occupied. Na is in Group 1, Period 3, so it has 1 outer electron and 3 shells. Cl is in Group 17, Period 3, with 7 outer electrons and 3 shells. (c) Na and K are both in Group 1, so each has exactly one outer-shell electron. This shared outer arrangement is the hidden rule that makes them behave alike, which solves the Lesson 1 puzzle of why elements far apart in atomic number can act almost identically. Cl has 7 outer electrons, one short of a full shell, so it is very reactive. Ar has a full outer shell of 8, so it is stable and unreactive and can be used safely in welding. Neighbours in the table can therefore be very different because their outer-shell electron numbers are very different.
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Part 3: Chemical Families and Trends

Using Lesson 4, (a) name the five chemical families you studied and give the group of each. (b) Describe the trend in reactivity down Group 1 (Li, Na, K) and down Group 17 (F, Cl, Br), using evidence from your Trend Detective Table. (c) Write a Claim-Evidence-Reasoning statement about how the reactivity of potassium compares with that of lithium, using only your table evidence and the idea that position predicts behaviour. Then say what your model could and could not yet explain at that point in the unit.	(a) Alkali metals (Group 1), alkaline earth metals (Group 2), halogens (Group 17), noble gases (Group 18) and transition elements (the block between Groups 2 and 13). (b) Down Group 1 reactivity increases: $\text{Li} < \text{Na} < \text{K}$. Down Group 17 reactivity decreases: $\text{F} > \text{Cl} > \text{Br}$. Our Trend Detective Table showed these directions, so trends within a group are directional and predictable. Real examples include potassium fertilisers used in the Rift Valley (Group 1) and chlorine used to treat water at Gigiri, Nairobi (Group 17). (c) Claim: Potassium is more reactive than lithium. Evidence: In our Trend Detective Table the reactivity ratings rose going down Group 1 from Li to Na to K, and K sits below Li. Reasoning: Elements in the same group show similar behaviour in a directional trend, so an element's position in the table predicts its behaviour, and the element lower down Group 1 is the more reactive. At that point our model showed WHERE the trends happen but could not yet explain WHY; we needed to look at electron arrangement as the next piece of evidence, which we then used with ions in Lesson 5.
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Part 4: From Electrons to Ions, Valency and Formulae

Using Lessons 5 to 7, (a) use dot-and-cross ideas to explain how Na and Cl form ions and why the ions are different. (b) State the oxidation numbers for Groups 1, 2, 16 and 17, and explain why iron at Kisumu Port can show Fe^{2+} and Fe^{3+}. (c) Use the crossover method to	(a) Na (2,8,1) loses its one outer electron to become Na^+ with 10 electrons (2,8), a stable arrangement. Cl (2,8,7) gains one electron to become Cl^- with 18 electrons (2,8,8). Na becomes a positive cation and Cl a negative anion because their electron transfers go in opposite directions. Atoms do this to reach a full outer shell, the octet or noble-gas arrangement; they do not 'want' electrons, they become more stable. (b) Group 1 is +1, Group 2 is +2, Group 16 is -2 and Group 17 is -1. These come from the number of electrons lost or gained to reach a stable arrangement. Iron is a transition element with an incompletely filled d sub-shell, so it can lose different numbers of electrons depending on conditions, giving Fe^{2+} or Fe^{3+} and different coloured compounds on rusted iron. (c) Calcium chloride: Ca^{2+} and Cl^-. Crossing the valencies gives Ca_1Cl_2, written CaCl_2. Sodium sulfate: Na^+ and SO_4^{2-}. Crossing gives $\text{Na}_2(\text{SO}_4)_1$, written Na_2SO_4; brackets are needed only if a radical has a subscript greater than 1. Check: in CaCl_2, $+2$ and $2 \times (-1) = 0$. In Na_2SO_4, $2 \times (+1)$ and $-2 = 0$. The total positive charge must equal the total negative charge so the
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derive the formulae of calcium chloride and sodium sulfate (radical SO_4^{2-}), and explain how you can check that each formula is correct.	compound is electrically neutral.
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Part 5: Balanced Equations and Answering the Driving Question

Using Lessons 8 to 10, (a) write and balance the equation for sodium reacting with water (products sodium hydroxide and hydrogen) and explain why you change coefficients and not subscripts. (b) Using sodium chloride from the Malindi salt flats, explain step by step how an element's position in the periodic table lets you predict its behaviour, its compound and its equation. (c) State one question that is still open and which sub-strand it leads to.	(a) Word equation: sodium + water $\rightarrow$ sodium hydroxide + hydrogen. Unbalanced: $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$. Atom inventory shows 2 H on the left and 3 H on the right (NaOH 1 + H_2 2), so balance: $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$. Now there are 2 Na, 4 H and 2 O on each side. Atoms are conserved in reactions, so each type of atom must be equal on both sides, in line with the Law of Conservation of Mass. We change coefficients because changing a subscript would change the substance itself (H_2O to H_2O_2 is a different substance). (b) Na is in Group 1, Period 3, so it has 3 shells and 1 outer electron. It loses that electron to form Na^+, with oxidation number +1. Cl is in Group 17 with 7 outer electrons, gains one to form Cl^-, oxidation number -1. Crossing the valencies gives NaCl, whose charges balance to zero. The reactions of such elements can then be written as balanced equations. This resolves the Lesson 1 puzzle: Na and Cl look completely different on their cards, yet their electron arrangements are complementary (one electron to give, one space to receive), so they combine into the salt that seasons ugali. The periodic table is a predictive map, not just a list. (c) An open question is: why do Na^+ and Cl^- attract each other, and what does that attraction mean for the properties of substances such as salt? This leads to Sub-Strand 1.4 on Chemical Bonding.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the pattern with electron arrangement	Clearly links group number to outer-shell electrons and period number to shells, and uses this to explain why same-group elements behave alike and neighbours differ, with correct examples such as Na/K and Cl/Ar.	States that group number relates to outer electrons and gives a mostly correct explanation, with limited examples.	Describes patterns on the table without linking to electron arrangement, or confuses groups and periods.
Use of evidence from the lessons	Uses specific evidence from several lessons (card sorting pairs, history, Trend Detective Table, dot-and-cross diagrams, balanced equations) and Kenyan contexts to support each claim.	Uses some specific evidence but is general in places or draws on only a few lessons.	Gives few or no specific pieces of evidence, or evidence does not match the claim.
Scientific accuracy of ions, valency and formulae	Correctly writes electron arrangements, ions, oxidation numbers, crossover formulae and a balanced equation, and checks charge balance and atom counts.	Mostly correct with one or two minor errors in notation, formulae or balancing.	Several errors, for example wrong ion charges, wrong formulae, or changing subscripts when balancing.

Reasoning and connection to the Driving Question	Builds a clear chain from position in the table to electrons, ions, valency, formula and equation, and uses it to resolve the Lesson 1 mystery and name an open question for Sub-Strand 1.4.	Links most steps in the chain but with some gaps in reasoning.	Lists facts without connecting them or without returning to the Driving Question.
Scientific vocabulary and communication	Uses terms such as group, period, valence electrons, cation, anion and oxidation number precisely, in clear complete sentences.	Uses most terms correctly with occasional unclear sentences.	Uses terms incorrectly or writes in fragments that are hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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